Reader Monad

Season: 7 · Advanced FP · Category: Functional · Level: Advanced

Intent

- Thread an environment through a computation.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Every function reads the same config clipboard.

When to Use

- Serious FP architecture in Java.
- The design force named in Intent shows up more than once.

When Not to Use

- CRUD apps that don't need effect algebras.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.reader;

import java.util.function.Function;

/** Reader monad — computation that depends on an environment R. */
public record Reader<R, A>(Function<R, A> run) {
    public A run(R env) { return run.apply(env); }

    public <B> Reader<R, B> map(Function<? super A, ? extends B> f) {
        return new Reader<>(env -> f.apply(run(env)));
    }

    public <B> Reader<R, B> flatMap(Function<? super A, Reader<R, B>> f) {
        return new Reader<>(env -> f.apply(run(env)).run(env));
    }

    public static <R, A> Reader<R, A> pure(A a) { return new Reader<>(env -> a); }
    public static <R> Reader<R, R> ask() { return new Reader<>(Function.identity()); }
}

record Config(String greeting, String name) {}
```

Modern Java 25 / Functional

- Witness types emulate higher-kinded types.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/reader/
```

Pitfalls

- Encodings are educational — production may use libraries.
- Tagless Final needs Kind encoding without HKTs.

Interview Q&A

- Q: When to pick Reader Monad? A: When you need testable effect boundaries or pure state threading.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Thread an environment through a computation.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Every function reads the same config clipboard. [0:40] Intent — State the design force in one senten...